

Architecture Document Template

Document title.

Document authors.

Document reviewers' and approval table

1. Table of contents (ToC)
2. Document scope
3. Introduction of the product and software
4. High-level, "20,000 feet" overview—a block diagram showing all the software blocks and the interfaces between them, the behaviour of the blocks, and the justification for arriving at this architecture.
5. Price points (the selling price range of the final product, along with the prices of competing products in the same market space).
6. Anticipated customers and customers' customers, if any (with references for the feasibility report and the requirements document).
7. A layered software architecture diagram.
8. A simple block diagram that shows where this software sits in the product.
9. A simple sketch to show where the product sits in the entire scheme of things. This could be a solutions architecture diagram, or a context diagram. For example, if you are creating a VoIP application, you sketch the product on which this will run, and the network in which the product will be deployed.
10. Operating system, hardware platform, frameworks and third-party tools used, with reasons for their usage.
11. 4+1 view, if applicable. (The 4+1 view means the five perspectives of product development—the design view, implementation view, process view, deployment view and use-case view. These have already been discussed in the previous article—Linux For You, April 2010 issue, Page 55.)
12. High-level diagram of interaction between the software components.
13. Stake-holders involved in this project.
14. Assumptions and constraints:
 - a. A list of non-functional requirements and their values. For example:
 - i) Response time should not exceed 250 ms.
 - ii) Minimum bandwidth expected is 2 MBps.
 - iii) The database should support at least 1 million users.
 - b. Any assumptions that we have to start with.
15. Estimated time and materials for software development, and the estimation methodology used (just the time, resources and the name of the technique need to be mentioned).
16. Standards impacted.
17. References.

the architecture diagrams. Unified Modeling Language (UML) is considered an Architecture Description Language (ADL). You can use UML to show the architecture, but it is only a guideline. Look at the *Architecture Document Template* box for a template for architecture documents.

Platform architect's role

Let's look at the role of a platform architect—in particular, a Linux platform architect. Platform architects in Linux will analyse, design, implement, support and manage the Linux/UNIX infrastructure for global customers. They will lead

Linux/UNIX support teams, and build and implement best practices on the Linux platform. They will have an excellent understanding of Linux platform integration in heterogeneous environments (like integration with Windows systems), and a deep understanding of the Linux core platform and system services such as NIS, NFS, DNS, etc. They will bring in a deep understanding of Linux fundamentals, including troubleshooting and debugging. They also have strong ideas relating to Linux vis-a-vis UNIX and Windows, and have a sound knowledge of different types of servers.

We will throw more light on these aspects in subsequent articles in this series.

In our next article, we will focus on how platform architects can choose a suitable operating system to meet their requirements. 

Resources

1. Software Architecture Consulting and Architect's Role <http://www.bredemeyer.com/DanaBredemeyer.htm>
2. Role of an Architect <http://www.sei.cmu.edu/architecture/research/competence/duties.cfm>
3. Software Architecture Definition and Role of an Architect www.engr.uconn.edu/~steve/Cse298300/umlbooch.ppt
4. 97 things every software architect should know—O'Reilly, Collective wisdom from the experts, edited by Richard Monson Haeffel, Shroff Publishers and Distributors Pvt Ltd.

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Gururajan currently works with Wipro Technologies as a lead consultant in the wireless and embedded domains. He has 16 years of core technology experience after a Masters degree in engineering, and has worked with various multinational companies in USA, Germany and India. Dr Thangaraju received a PhD in physics and worked as a research associate in IISc from 1996-2001. From 2001, he has been working in Wipro Technologies as a senior consultant and his core expertise is the Linux kernel, with expertise in embedded and real-time Linux.